

Beyond Semantic Confidence: Relation-Consistent Geometric Re-ranking for 3D Scene Graphs

Anonymous submission

Abstract

3D scene graph predictors can assign high scores to relation candidates whose predicates are inconsistent with the geometry of the reconstructed ordered object pair. We formulate predicate–geometry compatibility assessment for fixed relation predictions and introduce RelCompat3D, which preserves ordered-pair identity and separates predicate semantics, predicate-independent geometric measurements of the ordered pair, and the relation score produced by the predictor. We instantiate compatibility with family-specific linear heads and a compact shared nonlinear head. Both are trained using linked positive–counterfactual pairs without predictor identity or the predictor score. Transformation averaging enforces proximity symmetry and invariance under joint endpoint swap and inverse-predicate transformation. The method re-ranks proximity and vertical relations while keeping support/contact relations in source order. We evaluate Recall@ K with exact predicate matching jointly with Violation@ K , computed by a rule-based geometry verifier, across VL-SAT, Open3DSG, and a released SceneGraphFusion benchmark model (SGFN) on a shared 3DSSG target. Across the reported budgets, both RelCompat3D-Linear and RelCompat3D-MLP yield lower or tied Violation point estimates while preserving or improving Recall point estimates, with the largest gains on Open3DSG. Their different operating points, together with matched rank-fusion comparisons, reveal predictor- and family-dependent trade-offs.

Introduction

3D scene graphs represent reconstructed scenes as object–relation structures for spatial reasoning and embodied AI (Armeni et al. 2019; Wald et al. 2020; Sarkar et al. 2023; Liu et al. 2026). Their predicted relations must describe the corresponding ordered subject–object pair in 3D, not merely sound plausible at the category or language level.

We study high-scoring plausible predicates that conflict with measured ordered-pair geometry (Nguyen et al. 2026; Neau et al. 2026). Category-level plausibility can coexist with inconsistent distance, projected overlap, or vertical arrangement; contact-dependent relations are assessed separately by the verifier. This mismatch becomes consequential

as 3D scene graphs become interfaces for open-vocabulary language-based applications and embodied systems (Koch et al. 2024b; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025; Saxena and Chiun 2025; Yu et al. 2025; Madhavaram et al. 2026).

Figure 1 illustrates this failure.

The failure is not simply that 3DSSG methods (Wald et al. 2020) ignore geometry. Prior work already uses edge features, point-cloud structure, spatial knowledge, multimodal semantics, and geometric fusion (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Chen et al. 2024). The gap is more specific: a predictor score—even when its predictor already uses geometry—is not necessarily an explicit estimate of whether the same subject-object pair satisfies the predicate in 3D. This makes the problem an AI reliability issue, not only a 3DSSG implementation detail: language-based and embodied systems can inherit a high-scoring symbolic relation that contradicts the reconstructed scene evidence.

This failure suggests separating predicate semantics T , predicate-independent geometric measurements of the ordered pair G , and the predictor score Z . RelCompat3D preserves each candidate’s subject–object identity and learns predicate–geometry compatibility from T and G without using Z or predictor identity. We study a family-specific linear estimator and a compact shared nonlinear estimator under the same separation. Linked positive–counterfactual pairs optimize the ordering of geometrically compatible and incompatible candidates; averaging over known endpoint/predicate transformations enforces proximity symmetry and invariance under joint endpoint swap and inverse-predicate transformation. These choices address distinct failure modes: excluding Z precludes direct copying of the predictor score, preserving ordered-pair identity prevents scene-level measurements from substituting for instance evidence, and transformation averaging assigns equal compatibility to equivalent endpoint/predicate representations.

The predictor score enters only during re-ranking. Within proximity and vertical relations, RelCompat3D combines it with compatibility and changes the order within each relation family. It preserves the sequence of relation-family labels in the source ranking and the source order of support/contact candidates. Matched fusion baselines and ablations test the effects of excluding the source predictor score, preserving

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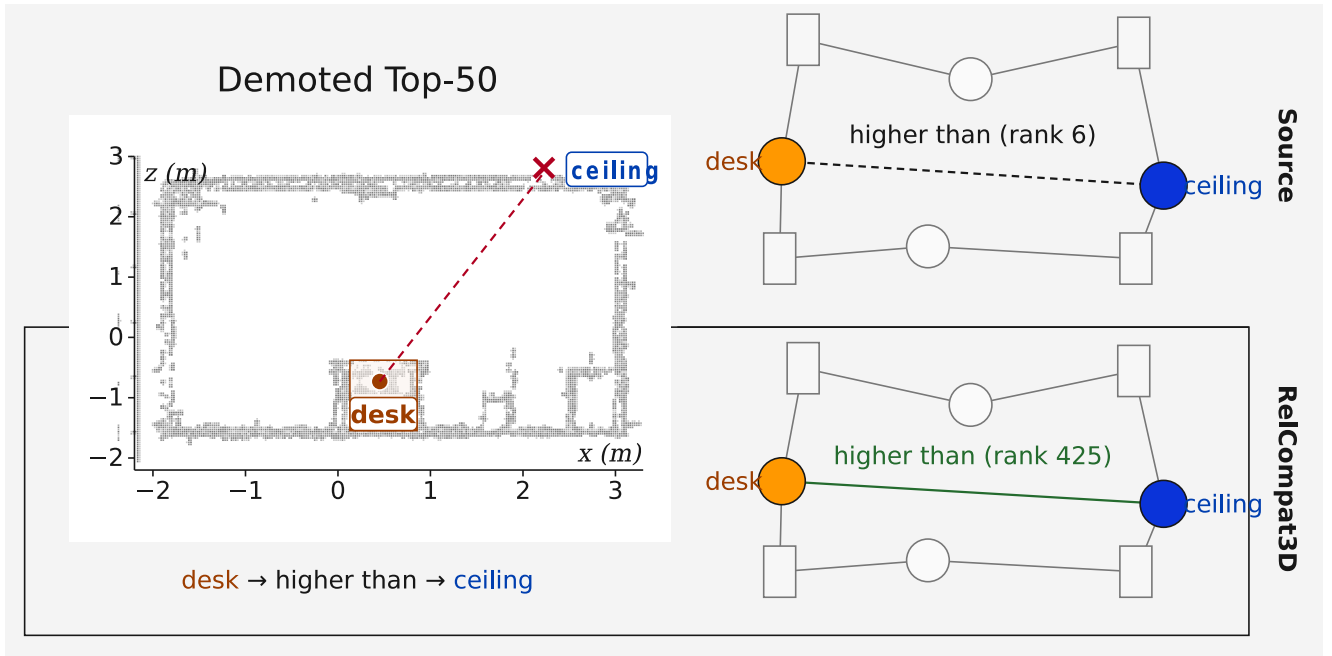


Figure 1: Open3DSG (Koch et al. 2024b) example. Measured elevation contradicts `desk higher than ceiling`; family-aware re-ranking moves it from rank 6 to 425, outside the Top-50.

ordered-pair identity, and enforcing transformation consistency. Figure 2 summarizes the framework.

We evaluate support/contact, proximity, and vertical-order relations because their consistency can be assessed from the geometry of reconstructed ordered object pairs. Open3DSG, VL-SAT, and the released SceneGraphFusion benchmark model (SGFN) provide complementary open-vocabulary and closed-set predictors (Koch et al. 2024b; Wang et al. 2023; Wu et al. 2021). We jointly report $\text{Recall}@K$ with exact predicate matching and $\text{Violation}@K$ computed by a rule-based geometry verifier across the three predictors on a shared 3DSSG target.

Our contributions are threefold:

1. We define and measure a mismatch between relation scores and instance-level geometric consistency that exact-match Recall alone does not capture, using identity-preserving evaluation and joint $\text{Recall}@K$ – $\text{Violation}@K$ reporting.
2. We learn predicate–geometry compatibility from the corresponding ordered pair without using the predictor score, and enforce exact endpoint/predicate transformations through transformation averaging.
3. We introduce a constrained re-ranking rule that preserves the source relation-family sequence and support/contact candidates in source order, and show that linear and non-linear compatibility estimators exhibit the same Source-level improvement pattern across three predictors while occupying different Recall–Violation operating points.

Across all reported predictor–budget settings, each compatibility estimator weakly improves the Source point-estimate Recall–Violation trade-off. At $K = 50$, paired in-

tervals from a cluster bootstrap over scans support lower Violation without a detectable Recall loss. RelCompat3D operates on fixed relation predictions rather than replacing the relation generator.

Related Work

3D Scene Graph Prediction

3D scene graphs organize reconstructed environments as object nodes connected by semantic and spatial relation edges (Armeni et al. 2019). 3RScan provides recurring indoor reconstructions, and 3DSSG adds object–relation annotations and a relation-prediction benchmark (Wald et al. 2019, 2020). Incremental and online systems study persistent graph construction from RGB-D streams and spatial perception pipelines (Wu et al. 2021; Hughes, Chang, and Carlone 2022). Recent predictors improve closed-set relations through edge reasoning, point-cloud features, multi-modal cues, spatial knowledge, self-supervised pretraining, and object- or edge-centric representations (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Heo et al. 2025; Ma et al. 2026). These methods and RelCompat3D both use reconstructed scene evidence, but they optimize relation generation, whereas RelCompat3D assesses and re-ranks fixed relation candidates after prediction.

Open-vocabulary 3D perception provides queryable object and region features (Peng et al. 2023; Takmaz et al. 2023), and graph systems connect such features to scene representations for querying, planning, navigation, online mapping, and language interaction (Gu et al. 2024; Werby et al. 2024; Maggio et al. 2024). Recent 3D scene graph methods extend this direction to open-vocabulary objects, open-set relations,

High-scoring relation contradicted by geometry. Compatibility estimation and Family-aware Re-ranking

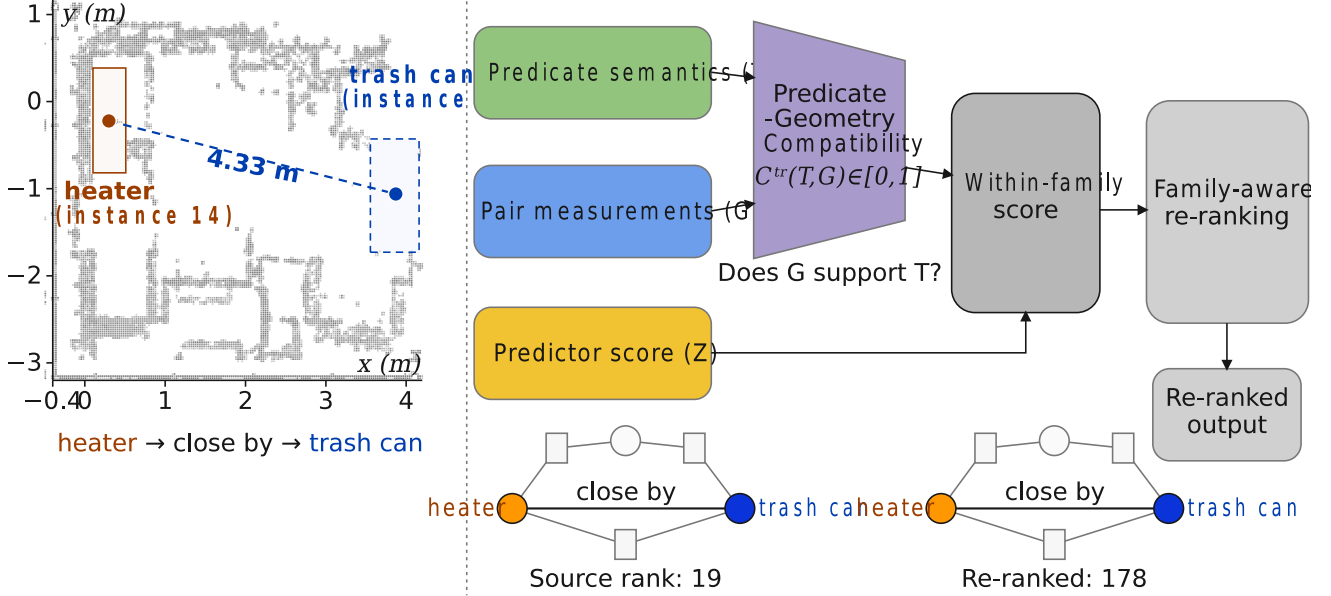


Figure 2: RelCompat3D overview. (a) Pair Geometry and Relation shows a high-scoring heater close by trash can prediction despite 4.33-m separation. (b) Compatibility and Re-Ranking computes $C^{tr}(T, G)$ from predicate semantics T and pair measurements G without predictor score Z , then combines Z only in the within-family score, moving the relation from rank 19 to 178.

vision-language model (VLM) features, online graph generation, and functional relations (Koch et al. 2024b; Chen et al. 2024; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025). These methods broaden the candidate vocabulary and downstream uses; RelCompat3D instead keeps each predictor fixed and tests whether its selected predicate is compatible with the reconstructed geometry of the corresponding ordered pair.

Geometry-Aware Relation Evidence

3D relation prediction and graph reuse already exploit edge features, metric distance, topology, support/contact patterns, and semantic-geometric fusion (Zhang et al. 2021; Feng et al. 2023; Xie, Pagani, and Stricker 2024; Huang et al. 2025; Shao et al. 2025; Sarkar et al. 2023). RelWitness learns open-vocabulary relations from visual-geometric witnesses under incomplete relation supervision (Nguyen et al. 2026). These approaches and RelCompat3D share the goal of grounding semantic relations in geometric evidence; RelCompat3D differs by excluding predictor identity and score from compatibility and applying the resulting signal to fixed candidates within each relation family.

Several systems incorporate geometry or transformation structure inside the relation generator. GEODE jointly denoises object geometry and predicates (Feng et al. 2026), RelGraphOV uses relation-aware graph construction and geometric pruning (Chen et al. 2026), and Transformation-Aware Decoupling improves generator robustness to yaw-viewpoint changes (Sun et al. 2026). For fixed 2D scene-graph-generation predictions, Neau et al. mine spatial, functional, and relation-algebra constraints and report a Con-

straint Violation Rate after declarative refinement (Neau et al. 2026). These methods and RelCompat3D all impose structured relation evidence, but RelCompat3D estimates continuous compatibility from the reconstructed ordered pair and enforces applicable endpoint/predicate transformations before joint Recall@ K -Violation@ K evaluation.

Reliability Evaluation and Calibration

Confidence calibration asks whether model scores reflect empirical accuracy. Temperature scaling adjusts neural confidence without changing the predicted class order (Guo et al. 2017); deep ensembles estimate predictive uncertainty, evaluation under distribution shift tests its robustness, and selective prediction trades conditional risk for coverage (Lakshminarayanan, Pritzel, and Blundell 2017; Ovadia et al. 2019; Geifman and El-Yaniv 2017). Scene-graph work models semantic ambiguity, calibrates long-tailed predictions, and constructs conformal prediction sets (Yang et al. 2021; Zhu et al. 2024; Nag et al. 2025). In 3D, SCR-SSG refines relation scores from masks, neighboring relations, and statistical priors, while PUF propagates uncertainty during online fusion (Yeo, Li, and Lee 2025; Yang et al. 2026). These methods and RelCompat3D both address reliability beyond task accuracy, but C^{tr} measures predicate-geometry compatibility rather than calibrated correctness, predictive uncertainty, or abstention confidence.

Recall@ K measures retrieval rank rather than compatibility with reconstructed ordered-pair geometry. RelCompat3D therefore reports exact-label Recall@ K alongside verifier-derived Violation@ K and uses compatibility for within-

family re-ranking. This joint evaluation complements conventional retrieval and calibration analyses; neither metric is a probability guarantee.

Method

RelCompat3D assesses fixed relation predictions rather than replacing a 3D Scene Graph generator. It preserves candidate identity, estimates predicate–geometry compatibility without the predictor score, and combines the two only during re-ranking (Figure 2).

Problem Setup

Let a relation predictor produce relation candidates $r_i = (\text{scan}_i, \text{context}_i, s_i, o_i, p_i, a_i, Z_i)$, where s_i and o_i are ordered subject and object instance identifiers, p_i is the predicate, a_i its mapped relation family, and Z_i the predictor’s native relation-ranking score. We use sigmoid relation scores for VL-SAT and SceneGraphFusion (SGFN), and normalized text-embedding cosine similarity for Open3DSG (Wang et al. 2023; Wu et al. 2021; Koch et al. 2024b), without predictor-specific normalization or refitting. Predictors are ranked separately; scores are never compared across predictors. A context is a scene subgraph over which top- K selection is performed independently. We distinguish the ordered-pair identity $k_i^{\text{pair}} = (\text{scan}_i, \text{context}_i, s_i, o_i)$ from the exact relation-candidate identity $k_i^{\text{rel}} = (\text{scan}_i, \text{context}_i, s_i, p_i, o_i)$ used for evaluation. The ordered-pair identity is retained when associating each prediction with geometry and ground truth. We represent predicate semantics as $T_i = p_i$. The family label a_i always selects the transformation set and re-ranking scope. It also selects the head and training-split normalization statistics for the linear estimator, whereas the shared nonlinear estimator encodes it as a semantic indicator. Each candidate is associated with predicate-independent ordered-pair measurements $G_i = G(\text{scan}_i, \text{context}_i, s_i, o_i)$. The reported evaluation scope is $\mathcal{A}_{\text{eval}} = \{\text{support/contact}, \text{proximity}, \text{vertical order}\}$, whose consistency can be assessed from the geometry of the reconstructed ordered object pair. Candidates outside these families are excluded from the reported Recall and Violation evaluation. The re-ranking scope is narrower: $\mathcal{A}_{\text{rank}} = \{\text{proximity}, \text{vertical order}\}$, for which the defined endpoint/predicate transformations are exact. Support/contact is evaluated but kept in source order because this family requires richer local contact and pose evidence, and no single endpoint transformation preserves the semantics of every predicate in the family.

For estimator $q \in \{\text{lin}, \text{mlp}\}$, our objective is to estimate a bounded compatibility score

$$C_i^q = \sigma(f_q(T_i, a_i, G_i)) \in [0, 1], \quad (1)$$

then average over the valid family-specific endpoint/predicate transformations to obtain the transformation-consistent score $C_i^{\text{tr}, q}$. The predictor score enters only through the within-family ranking score defined below. The compatibility and ranking scores are used for ordering and are not interpreted as posterior probabilities of physical validity.

Training-split ground truth provides positives. Proximity negatives use distant, non-overlapping same-context pairs; vertical negatives invert the predicate on the same pair; support/contact negatives replace one endpoint under separation margins. Separately, relation-preserving augmentation swaps proximity endpoints without changing the predicate and jointly swaps vertical endpoints and the inverse predicate. All training examples come from the training split; evaluation rows, predictor scores, and verifier-status labels are excluded, although the construction rules and primary verifier share some oriented-bounding-box (OBB)-derived measurements. Full construction rules are provided in the supplement.

Relation-Consistent Compatibility

For each candidate relation r_i , the associated geometry vector G_i contains OBB-derived 3D/XY distances, relative height, projected overlap, and vertical-gap features. Point-level contact evidence is used only by the support/contact verifier, not by the compatibility model. Let std_{tr} standardize each numeric feature with training-split statistics and replace a missing value by the corresponding training mean. We evaluate two compact compatibility estimators under the same input separation, training targets, loss, and transformation averaging.

Linear estimator. The feature map for RelCompat3D-Linear is

$$\begin{aligned} \Phi_{\text{lin}}(T_i, G_i) &= [1; \phi_T(T_i); \text{std}_{\text{tr}}(G_i); \text{std}_{\text{tr}}(\phi_{\times}(T_i, G_i))], \\ \phi_{\times}(T_i, G_i) &= [d(p_i)\Delta z_i, d(p_i)\Delta z_i^{\text{norm}}]. \end{aligned} \quad (2)$$

Here ϕ_T contains the predicate indicator, and $d(p_i)$ is 1 for higher than, -1 for lower than, and 0 otherwise; hence the two explicit interaction terms are active only for vertical-order predicates. The family label selects a linear head but is not repeated as a constant input: $f_{\text{lin}}(T_i, a_i, G_i) = w_{a_i}^T \Phi_{\text{lin}}(T_i, G_i)$.

Nonlinear estimator. RelCompat3D-MLP uses one shared two-hidden-unit ReLU network. Its input Ψ_i contains family and predicate indicators, the same underlying geometric measurements, the two signed-height interactions above, and the sum of the two directional overlap ratios. With a predicate-linear skip connection, its logit is

$$\begin{aligned} h_i &= \text{ReLU}(W\Psi_i + b), \\ f_{\text{mlp}}(T_i, a_i, G_i) &= v^T h_i + b_o + \beta^T \phi_T(T_i). \end{aligned} \quad (3)$$

The nonlinear estimator is shared across families, so its family indicator is not a constant feature. Both estimators exclude the predictor score, rank, predictor identity, and object-class features.

For proximity, τ swaps the ordered endpoints and leaves `close` by unchanged. For vertical order, τ swaps the endpoints and maps `higher than` $\leftrightarrow$ `lower than`. Transformation averaging therefore gives $C_{\text{close}}^{\text{tr}, q}(s, o) = C_{\text{close}}^{\text{tr}, q}(o, s)$ and $C_{\text{higher}}^{\text{tr}, q}(s, o) = C_{\text{lower}}^{\text{tr}, q}(o, s)$.

Training combines transformation-consistent augmentation with a linked positive–counterfactual ranking loss. The

augmentation preserves the predicate when proximity endpoints are swapped, whereas vertical order swaps endpoints and applies the inverse predicate. In addition, for every linked positive-counterfactual pair $(i^+, i^-) \in \mathcal{P}$, the logs $\ell_i^q = f_q(T_i, a_i, G_i)$ receive a fixed margin penalty:

$$\mathcal{L}_q = \mathcal{L}_{\text{BCE}} + 0.25 \mathbb{E}_{\mathcal{P}} \left[\log \left(1 + e^{1 - (\ell_{i^+}^q - \ell_{i^-}^q)} \right) \right] + 10^{-4} \mathcal{R}(\theta_q). \quad (4)$$

Here $\mathcal{L}_{\text{BCE}}$ is binary cross-entropy over positive and counterfactual examples. The second term is a soft margin-ranking loss with margin $m = 1$; its weight 0.25 and the ℓ_2 coefficient 10^{-4} are shared across estimators and predictors without predictor-specific search, and $\mathcal{R}$ penalizes non-bias weights. The supplement reports sensitivity, optimization details, and training-example counts. Both estimators are fitted exclusively on the training split.

Finally, let H_{a_i} denote the finite transformation group for family a_i , and define the transformation orbit $\mathcal{O}_i = \{g(T_i, G_i) : g \in H_{a_i}\}$. We average the learned score over this set to obtain the transformation-consistent compatibility:

$$C_i^{\text{tr}, q} = \frac{1}{|\mathcal{O}_i|} \sum_{(T', G') \in \mathcal{O}_i} \sigma(f_q(T', a_i, G')). \quad (5)$$

Group averaging makes proximity symmetry and invariance under joint endpoint swap and inverse-predicate transformation exact at inference. It does not use the predictor score, while the pairwise objective still orders each linked positive above its counterfactual. The supplement provides a formal statement and proof. For the Product (all families) comparison, $H_a = \{\text{id}\}$ for support/contact.

Family-Aware Re-ranking

For either compatibility estimator q , define the within-family ranking score

$$u_i^q = \begin{cases} Z_i C_i^{\text{tr}, q}, & a_i \in \mathcal{A}_{\text{rank}}, \\ Z_i, & a_i = \text{support/contact}. \end{cases} \quad (6)$$

For positive factors, $\log u_i^q = \log Z_i + \log C_i^{\text{tr}, q}$ in the families with defined transformations; compatibility contributes an additive log-score term without a fitted fusion parameter.

Let $\pi^Z = (e_1, \dots, e_n)$ denote the source ranking and a_j the relation family at position j . For each $a \in \mathcal{A}_{\text{rank}}$, let π_a^q contain that family's candidates sorted by u_i^q , with ties resolved deterministically by exact relation-candidate identity. For support/contact, π_a^q is defined directly as the corresponding subsequence of π^Z for both estimators. At position j , the rule selects the next unused candidate from the corresponding family list, exactly preserving the source family-label sequence. Its support/contact subsequence and family-specific Recall and Violation are therefore unchanged at every K . For each prefix, the rule also selects the highest-scoring candidates within each re-ranked family and maximizes their summed ranking score subject to the preserved family counts and support/contact subsequence; the supplement gives an exchange proof.

RelCompat3D-Linear and RelCompat3D-MLP share the factor separation, training targets, loss, transformations,

product ranking score, and family-aware ranking procedure; they differ in parameterization and nonlinear capacity. Rank-fusion baselines use the same ranking procedure with alternative score combinations. Additional controls test the predictor score, pair identity, geometry, and the defined transformations; full definitions are provided in the experiments and supplement.

Experiments

Experimental Setup

Data and splits. Predictions come from Open3DSG, VL-SAT, and the released SceneGraphFusion (SGFN) model SGFN_full_1160 (Koch et al. 2024b; Wang et al. 2023; Wu et al. 2021). We evaluate all three on one 3DSSG/3RScan target (Wald et al. 2019, 2020), restricted to relations whose consistency can be assessed from reconstructed ordered-pair geometry. All evaluations use the same scope: 157 scans, 548 relation contexts, and 3,972 exact-match ground-truth relations from support/contact, proximity, and vertical-order families. Ground-truth membership is used only when computing Recall@ K , never to filter or rank candidate relations.

Open3DSG uses the publicly released model and preprocessing pipeline. Every evaluation context is retained, with unavailable candidate lists treated as empty; coverage and denominator sensitivities are in the supplement.

Baselines and training. We evaluate the two compatibility estimators defined in Method with identical training rows, constructed targets, split, objective, and family-aware ranking procedure. Their fitted parameters are applied to all three predictors without predictor-specific refitting. Matched rank-average fusion (RankAvg) and reciprocal rank fusion (RRF) (Cormack, Clarke, and Büttcher 2009) combine predictor and linear-compatibility ranks. Product (all families) additionally applies the linear support/contact head, and a pooled linear model tests whether family-specific fitting is necessary. Controls perturb the predicate, pair identity, geometric measurements of the ordered pair, endpoint order, predictor score, or distance measurements; we apply the principal controls to both proposed estimators, with exact transformations in the supplement. Normalization, imputation, and model parameters use only 1,061 training scans; model design and the applicable relation families were selected on the 117-scan development split. The 157-scan evaluation split is never used for fitting. All comparisons across predictors concern this shared target.

Metrics and uncertainty. Let $L_K(c)$ be the selected top- K candidates in context c and Y_c its ground-truth relations. Both sets use exact relation identity (s_i, p_i, o_i) within c ; family mapping is used only for re-ranking and never for label matching. We report

$$\text{Recall@K} = \frac{\sum_c |L_K(c) \cap Y_c|}{\sum_c |Y_c|}. \quad (7)$$

Let $D_K \subseteq \bigcup_c L_K(c)$ contain selected candidates for which the rule-based geometry verifier returns satisfied, uncertain, or violated, and let N_s, N_u, N_v denote their respective

Predictor	Ranking rule	$K = 5$		$K = 10$		$K = 20$		$K = 50$		$K = 100$	
		R	V	R	V	R	V	R	V	R	V
Open3DSG	Source	3.42	52.05	9.87	32.89	19.89	20.99	40.43	13.87	51.11	12.42
	RelCompat3D-Linear	3.73	0.94	11.38	2.33	23.62	3.13	44.18	3.42	56.85	3.24
	RelCompat3D-MLP	3.70	4.95	11.78	4.97	24.67	4.56	46.70	4.13	59.89	3.71
	RankAvg	3.78	1.13	11.05	2.51	22.36	3.55	43.20	4.60	54.00	5.49
	RRF	3.58	19.59	10.67	16.47	21.58	13.60	42.45	9.60	54.68	7.80
	Product (all families)	9.94	3.98	19.64	4.54	32.80	4.37	46.42	2.69	60.05	3.31
VL-SAT	Source	41.94	0.29	63.22	0.82	80.74	1.42	92.72	2.68	96.35	4.76
	RelCompat3D-Linear	42.07	0.15	63.39	0.57	80.82	1.14	92.77	1.97	96.58	2.95
	RelCompat3D-MLP	42.09	0.15	63.47	0.51	80.92	1.09	92.72	1.89	96.50	2.96
	RankAvg	35.32	0.15	51.74	0.47	69.41	0.99	90.23	1.62	97.05	2.26
	RRF	36.81	0.15	55.66	0.51	75.08	1.07	91.72	1.82	97.05	2.50
	Product (all families)	41.74	0.11	63.32	0.49	80.94	1.10	92.93	2.03	96.88	3.25
SGFN	Source	31.17	2.37	39.75	3.49	49.12	3.22	74.02	3.85	92.35	6.30
	RelCompat3D-Linear	31.17	2.37	39.75	3.47	49.14	2.97	74.50	2.63	93.03	3.50
	RelCompat3D-MLP	31.17	2.37	39.75	3.47	49.19	2.96	74.57	2.58	92.88	3.50
	RankAvg	31.09	2.37	38.80	3.47	46.63	2.95	70.22	2.35	92.72	2.60
	RRF	31.14	2.37	38.95	3.47	47.08	2.97	68.50	2.54	87.99	3.07
	Product (all families)	31.07	2.48	40.56	3.19	51.18	2.76	77.04	2.58	94.13	3.71

Table 1: Shared 3DSSG validation results. Rows group predictors and ranking rules; columns give exact-match Recall ($R\uparrow$) and verifier-derived Violation ($V\downarrow$) at each K (%). Source is the original ranking, RankAvg/RRF are matched fusion baselines, and Product (all families) applies linear compatibility to every family.

counts. The primary verifier-derived metric is

$$\text{Violation@K} = \frac{N_v}{N_s + N_u + N_v}. \quad (8)$$

Thus uncertain candidates enter the denominator but not the numerator. We also report uncertainty rate $N_u/|D_K|$, decidable-only violation $N_v/(N_s + N_v)$, status coverage $|D_K|/\sum_c |L_K(c)|$, and decidable coverage $(N_s + N_v)/|D_K|$; a pessimistic variant counts uncertain candidates as violations. We report $K \in \{5, 10, 20, 50, 100\}$ and discuss $K = 50$ as an intermediate reported budget. Paired 95% confidence intervals use 1,000 bootstrap resamples of the 157 scans, keeping all contexts from each scan together and using the same resamples across conditions. Within-family results and family-specific metrics within the selected top- K predictions show whether aggregate changes are driven by relation-family composition.

Recall–Violation Results

Table 1 and Figure 3 show that neither RelCompat3D variant lowers Recall nor raises Violation relative to Source in any of the 15 predictor–budget cells. At $K = 50$, scan-cluster intervals are positive for Recall and negative for Violation on Open3DSG and SGFN; VL-SAT Recall intervals contain zero while its Violation intervals remain negative. Open3DSG exhibits the largest gains, whereas SGFN changes only marginally at $K = 10$.

No single comparator dominates across all predictors and reported K values. At $K = 50$, RelCompat3D-MLP raises Open3DSG Recall more than RelCompat3D-Linear (46.70% versus 44.18%) but also has higher Violation (4.13% versus 3.42%); the two are nearly tied on VL-SAT and SGFN. Thus capacity changes the operating point without identifying a

universally superior head. The product variants avoid larger small- K Recall losses from rank fusion on VL-SAT and SGFN. Product (all families) can raise aggregate Recall but changes support/contact selections and can mask a family regression; matched variants preserve the source family sequence and support/contact order.

Surface-Based Geometry Audit

Table 3 remeasures proximity and vertical relations using point–mesh consensus with train-only thresholds and no OBB inputs or verifier labels. The Linear variant lowers $K = 50$ Surface-V across all predictors with intervals below zero; MLP and all- K audits are supplemental.

Ablations and Controls

Table 2 shows that wrong-pair and shuffled geometry lower Recall and raise Violation at $K = 50$. Wrong-predicate and fixed-label swap sharply increase Violation, while distance-only and compatibility-only ordering expose source-dependent trade-offs. Matched $K = 100$ controls, feature removal, linked-pair ordering, and transformation checks are in the supplement.

Discussion and Limitations

All three predictors share one 3DSSG target (Wald et al. 2020), and ReplicaSSG/FROSS (Hou et al. 2025) exhibits non-uniform transfer under ontology and geometry shifts (supplement). RelCompat3D assumes known instances and measurable ordered-pair geometry; support/contact errors, reference-frame-dependent relations, and complete graph generation remain outside scope. The compatibility targets

Control	Head	Open3DSG	VL-SAT	SGFN
Full method	Linear	44.18/3.42	92.77/1.97	74.50/2.63
	MLP	46.70/4.13	92.72/1.89	74.57/2.58
Wrong predicate	Linear	42.65/21.98	90.46/4.97	71.58/9.42
	MLP	46.78/19.30	90.99/5.03	71.78/9.67
Wrong pair	Linear	38.52/8.50	90.56/2.61	71.55/3.99
	MLP	38.02/9.31	91.31/2.62	72.10/3.91
Shuffled geometry	Linear	38.42/12.93	89.17/3.05	70.59/4.73
	MLP	36.83/12.88	90.26/3.00	71.15/4.59
Fixed-label swap	Linear	42.67/21.98	90.46/4.97	71.58/9.42
	MLP	45.52/19.13	90.99/4.99	71.85/9.65
Distance only	Both	51.16/8.24	81.90/5.34	63.19/10.00
Compatibility only	Linear	42.22/3.42	67.65/1.61	52.79/2.32
	MLP	48.19/3.42	77.84/1.61	57.10/2.33

Table 2: Matched $K = 50$ controls on the shared 3DSSG target. Cells give exact-match Recall / verifier-derived Violation (% , $\uparrow$ / $\downarrow$) by predictor, control, and head; “Both” denotes the shared distance-only control. $K = 100$ results are supplemental.

Predictor	Source	Ours	Δ [95% CI]	Cov.
VL-SAT	16.43	14.34	-2.09 [-2.41, -1.80]	95.5/71.0
Open3DSG	45.96	4.90	-41.06 [-42.75, -39.28]	98.2/83.7
SGFN	11.56	8.13	-3.43 [-3.92, -3.00]	95.8/79.9

Table 3: Surface-based Violation at $K = 50$ (% , $\downarrow$) from point-mesh consensus. Ours denotes RelCompat3D-Linear; Δ is Ours minus Source with a paired scan-cluster 95% CI, and Cov. is measured/decidable coverage. MLP and all- K results are supplemental. Values are not directly comparable to Table 1 V.

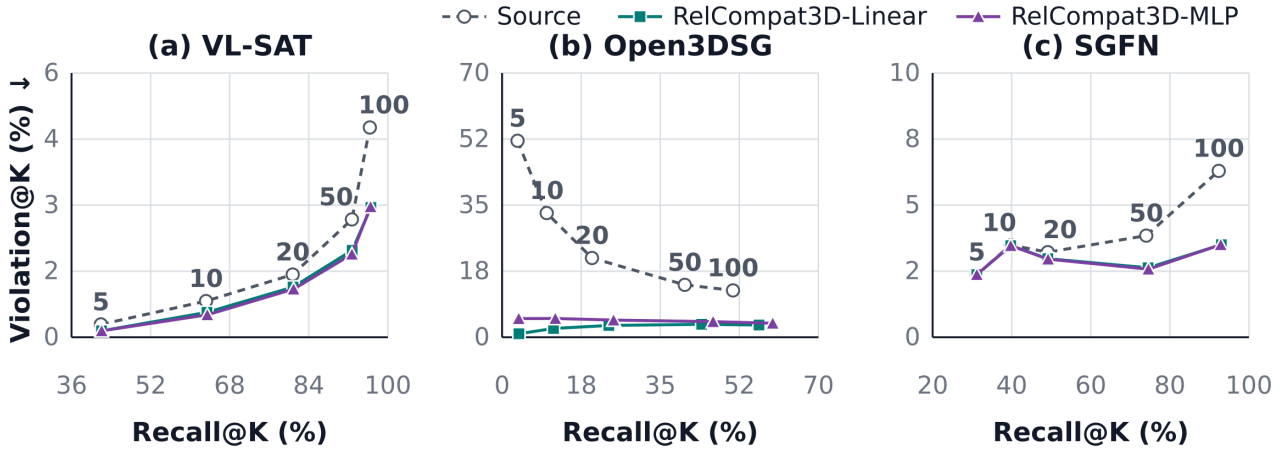


Figure 3: Recall@K–Violation@K trajectories. Each line connects $K \in \{5, 10, 20, 50, 100\}$ for Source, RelCompat3D-Linear, and RelCompat3D-MLP. Rightward and downward movement is preferred. Numbers label K along the Source trajectory; all curves follow the same budget order. Axis ranges differ by predictor.

and primary verifier share some OBB-derived measurements. The surface audit excludes these inputs and labels but shares the reconstructed surface and ontology, so it is not independent physical-validity evidence. Broader claims require independent labels, contact evidence, and datasets.

Conclusion

RelCompat3D separates predicate–geometry compatibility from predictor scores. Across three predictors on one shared target, both variants yield no worse reported Recall–Violation point estimates and do not re-rank support/contact.

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